

Instructor: Date:

Learner: Regional Branch:

Program: Center:

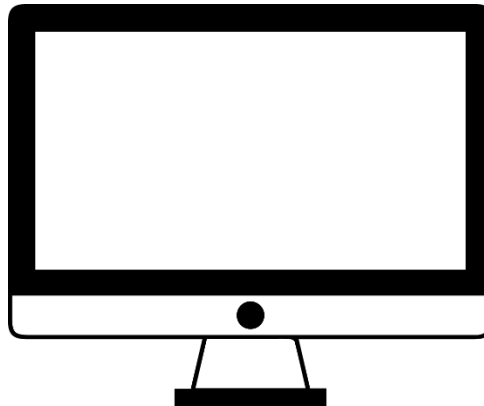
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Let's Start!

Getting Acquainted with Supercomputers.



Learning Outcome:

Identify key information about people's jobs and places of work in spoken or written contexts.
Make the information system design report, based on the selection of both hardware and software tools required for the IT solution.

Can Do Statements:

- I can explain what a supercomputer is in a written way.
- I can talk about the evolution of supercomputers throughout history.
- I can read about engineers who want to create a supercomputer.

Ask your partner:

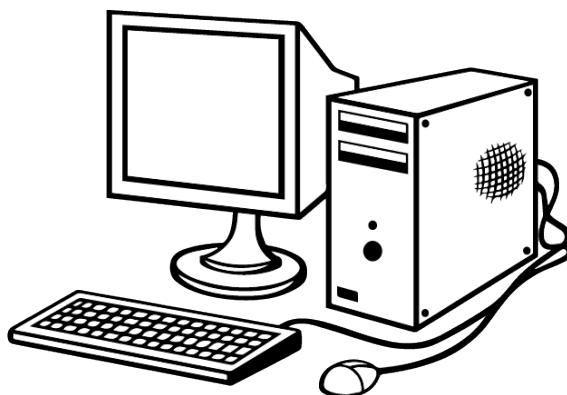
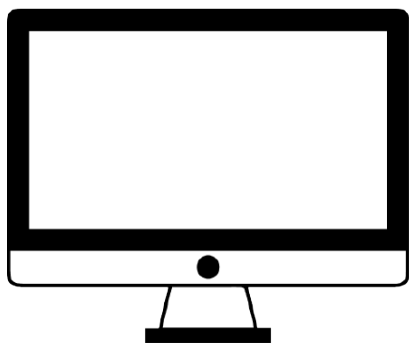
- Why do you think speed is an important factor when talking about computers?
- What is the importance of supercomputers in other fields?





Let's Explore!

Let's learn about the supercomputer!



Instructions: Read the text below.

Voice off: The development of the computer experienced important changes since the XXth century to this day. Such was the case that experts created the term "Supercomputer" to make reference to the new, and each time better, computer versions. In others words, a supercomputer is the fastest and most powerful machine model created for a specific goal in the technology field. This one must keep up to date with the most recent enhancements in computer architecture, such as the Sunway TaihuLight, which is the fastest supercomputer at present.

Now then, regarding the evolution of supercomputers, the first supercomputer ever mentioned was the Atlas, back in the 60's. The electrical engineer Seymour Cray designed it in 1962, but later in 1964, he had enough experience when he conceived the Data CDC 6600, which had silicon transistors and ran at 10 MHz. It was popular but not enough as the Cray I was, which soon became a perfect replacement for the CDC 6600 in 1976. This one had a more modern architecture, integrated circuits, and a vector processing. However, regarding parallelism, this was less functional than the next supercomputer, the Cray X-MP. This new version was better in terms of parallelism thanks to the incorporation of multiple processors.

On the other hand, supercomputers were less common in the market since these were more expensive than the usual ones. The reason is that Supercomputers are only used for special and advanced purposes. Consequently, engineers are constantly creating new inventions in order to improve the existing versions and offer solutions to any challenge that science and new generations impose. As supercomputers change almost every day, it is not enough for engineers; so, they have to develop new machines that can help with aircraft prototypes, the analysis of nuclear effects, the weather models, and other important projects, such as the National Security Agency or NSA that controls communication all around the world.



**Let's Practice!**

Angela is reviewing what she listened.

Instructions: Write the words to correctly complete the sentences. Use words from the box below.

had / Conceived the / referred

1. Seymour Cray _____ first supercomputer versions: Atlas, Data CDC 6600, Cray I, and Cray X-MP.
2. In terms of parallelism, the Cray X-MP _____ better features.
3. The term supercomputer _____ to the latest and fastest computer version in the XXth century.

**Let's Practice!**

Angela reads about supercomputers.

Instructions: Use the words from the Word Bank to complete each sentence.

Word Bank

less / are not enough / enough

1. For the science development, the existing supercomputer versions _____; so, engineers have to improve them.
2. Engineers are not creating _____ computers to cover the diverse areas of communication, aeronautics, and science, among others.
3. Engineers design _____ powerful computers for the public because the costs of supercomputers are very high.



Extension Activity

Supercomputers in your community.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for an industry in your community that might need a supercomputer.
- Write a paragraph where you explain why a supercomputer is important for that industry.
- Share your findings with a partner.

Self-Assessment

Yes

No

1. I search for an industry in my community that might need a supercomputer.

☐
☐

2. I create a paragraph where I explain why a supercomputer is important for that industry.

☐
☐

3. I share my findings with a partner.

☐
☐


Target Words

- **silicon** (noun): element found in rocks and minerals commonly used in the computer chips manufacture
e.g. but later in 1964, he had enough experience when he conceived the Data CDC 6600, which had silicon transistors and ran at 10 MHz.
- **vector** (noun): an internal organization of data to be found easily
e.g. This one had a more modern architecture, integrated circuits, and a vector processing.
- **parallelism** (noun): related to computing operations performed by a computer at the same time
e.g. This new version was better in terms of parallelism thanks to the incorporation of multiple processors.
- **impose** (verb): to require
e.g. Engineers are constantly creating new inventions in order to improve the existing version and offer solutions to any challenge that science and new generations impose.
- **aircraft** (noun): an air transportation mean, such as an airplane
e.g. They have to develop new machines that can help with aircraft prototypes, the analysis of nuclear effects, the weather models, and other important projects.



Learning Points

Quantifiers, Enough, Not Enough and Less.

You use **quantifiers** to describe the amount of something. Some common quantifiers are: **enough**, **not enough**, and **less**.

ENOUGH

You use **enough** to indicate that the corresponding amount is satisfactory.

Example:

- *Cray had **enough** experience when he conceived the Data CDC 6600 supercomputer.*

NOT ENOUGH

You use **not enough** to indicate that the corresponding amount is not satisfactory because you need more.

Example:

- *Advances in technology are **not enough** for computing scientists, so they continue seeking new challenges.*

LESS

You use **less** to indicate that the corresponding amount is smaller than the expected. Use **less** to make inferiority comparisons.

Example:

- *Windows Vista operating system was **less** popular than Windows XP.*

Examples

- *Seymour Cray had enough experience when he conceived the Data CDC 6600, which had silicon transistors and run at 10 MHz.*
- *It was popular but not enough as the Cray I, which was also conceived by Cray in 1976.*
- *However, regarding parallelism, this was less functional than the next supercomputer, the Cray X-MP.*
- *Supercomputers change almost every day, it is not enough for engineers, so they create new machines.*





Learning Points

Timeline, The Supercomputer.

Technology has the responsibility to help solving complex tasks in different knowledge domains. For this purpose, engineers work very hard designing supercomputers, mighty machines with special features to perform these tasks.

In this timeline, you can check the first supercomputers appearance as well as their gradual progress and improvements up to these days:

